Fazeel Nizam

+94768114000 nizamfazeel@gmail.com Kalutara, Sri Lanka
https://fazeelnizam.github.io/portfolio/ FazeelNizam
fazeelnizam Bē fazeelnizam

Profile

Innovative and self-driven Computer Engineering undergraduate with hands-on experience in embedded systems, IoT, FPGA/VHDL, and full-stack development. Proven ability to architect and build end-to-end systems, from firmware and hardware integration to data visualization dashboards and cloud connectivity. Strong collaborator and quick learner, seeking opportunities to design cutting-edge IoT, Software, and FPGA systems that bridge research and real-world applications.

Education

BSc (Hons) in Computer Engineering, *Open University of Sri Lanka* 2021 – present
GPA - 3.58

G.C.E A/L (Physical Stream), *Prince Of Wales College Moratuwa* 2017 – 2019

Technical Skills

Programming & Scripting, *C, Embedded C/C++, Python, Java, JavaScript (ES6), VHDL*

IoT, Hardware & Protocols, *ESP32, Arduino, Raspberry Pi, SDI-12, SPI/I^{2}C/UART/LoRa/BLE, JSON, MQTT*

Frameworks, *ReactJS, NextJS, ExpressJS, Django (IoT), Flask (IoT)*

Markup Language and Styling, *HTML5, CSS, SCSS, Tailwind, MUI*

Technologies and Tools, *Git, Figma, MATLAB, Proteus, AutoCAD, Vivado, Blynk IoT, AWS IoT, NodeRed*

Professional Experience

Technical Assistant, *Career Guidance Unit, OUSL* Mar 2025 – Sep 2025
Nawala, Sri Lanka

- Engineered a **Multi-Unit Environmental & Agricultural Monitoring System** using three wirelessly coordinated ESP32 nodes for real-time aggregation of soil and environmental data.
- Architected and developed a **Portable Greenhouse Gas (GHG) Analyzer** for agricultural field research with laboratory-grade accuracy in outdoor conditions.
- Provided technical support for university-wide career guidance events.
- **Technologies:** ESP32, Pi Zero, Embedded C, Python
- GitHub Repositories:
  Multi-Unit Environmental Monitoring System
  Portable Greenhouse Gas (GHG) Analyzer

Trainee Electronic Engineer, *Arthur C Clarke Institute for Modern Technologies* — Jul 2024 – Dec 2024, Katubedda, Sri Lanka

- Designed and implemented an **ESP32-based IoT Nano-Sterlite Orbital Magnetic Field Simulation Device**.
- Designed and implemented wireless data logging and remote control features via a web dashboard hosted in the ESP32.
- **Technologies:** ESP32, Embedded C, FreeRTOS, SPIFFS, Python, HTML, CSS

Technical Assistant, *Career Guidance Unit, OUSL* — Feb 2024 – Jun 2024, Nawala, Sri Lanka

- Assisted in managing and organizing university career programs
- Provided technical support for online and in-person events

Technical Assistant, *Neo Space Lab, OUSL* — Sep 2023 – Dec 2023, Nawala, Sri Lanka

- Designed UI/UX for a **Research Publication Web App** and **IoT dashboards** using NextJS and MUI, improving usability and responsiveness.
- **Technologies:** NextJS, ReactJS, HTML, CSS, MUI, Figma
- GitHub Repositories:
  IoT Weight Measurement System
  LKO Web Dashboard

Banking Trainee, *Commercial Bank of Ceylon, Main Branch Wellawatta* — Aug 2022 – Dec 2022

- Handled lower-counter operations, including account opening, debit card issuance, and E-Remittances, ensuring compliance and customer satisfaction

Featured Projects

Multi-Unit Environmental & Agricultural Monitoring System (On Going), — 2025
ESP32 | Pi Zero | Embedded C | Python | TEROS 10 | TEROS 22 | ATMOS 41

- Architected a distributed system with three wireless ESP32 nodes for real-time environmental and soil monitoring.
- Designed a Wi-Fi–based data pipeline where one node serves sensor data via a JSON API and another consumes and uploads data to the cloud via Pi Zero MQTT Broker.
- Programmed a self-sufficient weather station capable of parsing complex SDI-12 protocols and logging timestamped data.

Portable GHG Analyzer for Agricultural Field Research, — 2025
ESP32 | Embedded C

- Built a standalone, keypad-operated device for precision GHG measurement, improving field research efficiency.
- Implemented a dual-file system: SPIFFS for persistent calibration data and SD card for logging individual sample data separately.
- Developed automated control for air circulation and stabilization to ensure high-fidelity readings.

Helmholtz Cage - Nano Sterlite Orbital Magnetic Field Simulation Device, — 2024
Embedded C | ESP32 | Python | HTML | CSS

- Implemented IoT power control with a web dashboard hosted in ESP32 for remote monitoring
- Developed a self-balancing, frictionless air-bearing platform using PID algorithm
- Integrated and managed industry-level sensors and actuators

Social Learning Platform - Realtime Chat App (University Project), 2024
ReactJS | NodeJS | Stream.io
- Integrated Stream.io API for real-time chat functionality
- Designed and customized user interface for enhanced user experience

WPCU - Water Pump Control Unit (University Project), 2024
FPGA | VHDL | Vivado
- Designed and implemented an FSM-based control system for efficient water management
- Developed and tested the system on an FPGA development board

Smart Water Management System (University Project), 2024
ESP32 | Embedded C | FreeRTOS | LoRa | AWS IoT | MQTT | Blynk IoT
- Built a distributed IoT water-monitoring network with solar-powered LoRa sub-stations and a Wi-Fi base station for secure AWS IoT cloud integration.
- Developed FreeRTOS-based firmware to handle LoRa reception, cloud updates, and dashboard communication with real-time, non-blocking alerts.
- Streamed AES-encrypted sensor data via MQTT to AWS IoT and delivered a live Blynk mobile dashboard for remote data visualization.

IoT Weight Measurement System Dashboard, *NextJS | CSS | MUI* 2023
- Designed a real-time IoT data visualization dashboard
- Integrated backend API using Axios for data retrieval

Soft Skills

Leadership and Team Management, *Led multiple student organizations and University events*

Strong Interpersonal & Communication Skills, *Collaborated effectively with diverse teams*

Teaching and Mentorship, *Experience guiding peers and students in both technical and non-technical domains*

Multilingual, *Fluent in Sinhala, English, and Tamil*

Extra Curricular Activities

Student Volunteer, *USBus Mobile American Space Program 2025*

Organizing Committee Member, *TalentLyft '25 Career Fair, OUSL*

President, *ENOSPIRE OUSL 2025 Entrepreneurship Day Organizing Committee*

Technical Team Lead, *TalentLyft '23 Career Fair, OUSL*

Volunteer Teacher & Committee Member, *SVP Educational Institute, Kalutara*

References

Prof. Channa J Kumarage, *Computer Information Systems Professor,*
Olympic College, 1600 Chester Ave, Bremerton WA
cjkumarage@gmail.com, +1(320)2810271

Eng. D. S. Wickramasinghe, *Senior Lecturer,*
Department of Electrical and Computer Engineering, The Open University of Sri Lanka
dswic@ou.ac.lk, +94701372644